Lab 6 - Snort (Intrusion Detection & Prevention)

In this lab, we explore **Snort**, one of the world's most widely used Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) (owned by Cisco).

Snort can analyze network traffic in real time and detect a wide range of attacks, intrusions, exploits, and suspicious behaviors.

Snort operates in three modes:

1. **Sniffer Mode**: displays raw packet headers.
2. **Packet Logger Mode**: logs packets to disk.
3. **Network Intrusion Detection System (NIDS) Mode**: uses rules to detect attacks.

Answer the following questions and send the report to: <csclass.dz@gmail.com>
Team information (one report by team):

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- First Name Last Name: Zerouli Abd Assamaed

Deadline: Wednesday, Dec 17, 2025.

Check if Snort is installed by default on your Kali machine. If not, install it (type installation commands).

which snort
/usr/sbin/snort

```
└─$ which snort
/usr/sbin/snort
```

Report how you solved the issues related to Snort installation (used commands).
i got no problem when u install snort i just do

```
└─$ sudo apt install snort -y
```

Do not scan or attack systems you do not own (or have explicit permission to test). Unauthorized sniffing, probing, or intrusion of networks, IP addresses, or domains is illegal and may result in criminal or civil penalties.

Part 1 — Getting Started with Snort

1. Check the Snort version

Run on your Kali machine:

`snort -V` or `snort --version`

```
└─$ snort -V

   ,,_     -*> Snort++ <*-
  o"  )~   Version 3.10.0
   ''''    By Martin Roesch & The Snort Team
           http://snort.org/contact#team
           Copyright (C) 2014-2025 Cisco and/or its affiliates. Al
l rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using DAQ version 3.0.23
           Using libpcap version 1.10.5 (with TPACKET_V3)
           Using LuaJIT version 2.1.1737090214
           Using LZMA version 5.8.1
           Using OpenSSL 3.5.4 30 Sep 2025
           Using PCRE2 version 10.46 2025-08-27
           Using ZLIB version 1.3.1
```

Questions

- What version of Snort is installed?
  3.10.0
- Why is it important to keep IDS/IPS tools updated?
  New attacks and malware appear all the time, and updates allow the IDS/IPS to detect them.

  Updates add new rules and signatures needed to recognize recent threats.

  They help reduce false alarms and improve detection accuracy.

  Updates also fix bugs and security issues in the tool itself

2. Sniffer Mode

Run snort in sniffer mode: sudo snort -v - i (your interface)

```
└─$ sudo snort -v
[sudo] password for ababil:
Running in packet dump mode

        --== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "wlan0".
Decoding Ethernet

        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.20 GRE (Build 82)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.10.5 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.3.1

Commencing packet processing (pid=651817)
WARNING: No preprocessors configured for policy 0.
12/14-21:43:10.350204 190.2.149.6:443 -> 192.168.1.12:51133
UDP TTL:50 TOS:0x28 ID:63998 IpLen:20 DgmLen:140
Len: 112
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-21:43:10.364029 192.168.1.12:51133 -> 190.2.149.6:443
UDP TTL:64 TOS:0x0 ID:47109 IpLen:20 DgmLen:124
Len: 96
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-21:43:10.534014 192.168.1.12:51133 -> 190.2.149.6:443
UDP TTL:64 TOS:0x0 ID:47111 IpLen:20 DgmLen:124
Len: 96
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-21:43:10.588441 190.2.149.6:443 -> 192.168.1.12:51133
UDP TTL:50 TOS:0x28 ID:64044 IpLen:20 DgmLen:124
Len: 96
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```

Questions

- What kind of information does Snort show in this mode?
  Source IP address

  Destination IP address

  Source and destination ports

  Protocol used (TCP, UDP, ICMP)

  Packet headers and basic packet details

- How could this be useful for an attacker?
  The attacker can see **IP addresses** of active machines on the network.

  They can identify **open ports and services** being used

They can learn which **protocols** are commonly used

They can understand normal traffic patterns to plan attacks and avoid detection

- Why is it insufficient as a full IDS?
  It only **displays network traffic** and does not analyze it.

  It does **not use rules or signatures** to detect attacks.

  It does **not generate alerts** for malicious activity.

  It cannot block or prevent attacks.

3. Packet Logger Mode

Create a directory for logs: `mkdir /tmp/snortlog`
Run snort in packet logger mode: sudo snort -dev -l /tmp/snortlog

```
└─$ sudo snort -dev -l /tmp/snortlog

[sudo] password for ababil:
Running in packet logging mode

        --== Initializing Snort ==--
Initializing Output Plugins!
Log directory = /tmp/snortlog
pcap DAQ configured to passive.
Acquiring network traffic from "wlan0".
Decoding Ethernet

        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.20 GRE (Build 82)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.10.5 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.3.1

Commencing packet processing (pid=698530)
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
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WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
```

Then from your **Metasploitable**, ping the Kali machine.

Stop Snort.

Questions

- Which files were created in `/tmp/snortlog?`
-
```
└─$ ls /tmp/snortlog
snort.log.1765750800
```
- What information can an incident responder recover from these logs?
  source and destination **IP addresses**

  Source and destination **ports**

  Protocols used (TCP, UDP, ICMP)

  Timestamps of network communication

Payload data (depending on logging options)

Communication sessions between hosts

```
└─$ sudo snort -r /tmp/snortlog/snort.log.1765750800
Running in packet dump mode

        --== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "/tmp/snortlog/snort.log.1765750800".

        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.20 GRE (Build 82)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.10.5 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.3.1

Commencing packet processing (pid=703725)
WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.291760 192.168.1.12:51133 -> 190.2.149.6:443
UDP TTL:64 TOS:0x0 ID:53042 IpLen:20 DgmLen:1340
Len: 1312
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.291814 192.168.1.12:51133 -> 190.2.149.6:443
UDP TTL:64 TOS:0x0 ID:53043 IpLen:20 DgmLen:1340
Len: 1312
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.291835 192.168.1.12:51133 -> 190.2.149.6:443
UDP TTL:64 TOS:0x0 ID:53044 IpLen:20 DgmLen:1340
Len: 1312
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.291851 192.168.1.12:51133 -> 190.2.149.6:443
UDP TTL:64 TOS:0x0 ID:53045 IpLen:20 DgmLen:1340
Len: 1312
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.291975 192.168.1.12:51133 -> 190.2.149.6:443
UDP TTL:64 TOS:0x0 ID:53046 IpLen:20 DgmLen:876
Len: 848
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.363670 190.2.149.6:443 -> 192.168.1.12:51133
UDP TTL:50 TOS:0x28 ID:21651 IpLen:20 DgmLen:124
Len: 96
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.364812 190.2.149.6:443 -> 192.168.1.12:51133
UDP TTL:50 TOS:0x28 ID:21652 IpLen:20 DgmLen:796
Len: 768
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
12/14-23:20:02.364813 190.2.149.6:443 -> 192.168.1.12:51133
UDP TTL:50 TOS:0x28 ID:21653 IpLen:20 DgmLen:124
Len: 96
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```

Part 2 — Running Snort in IDS Mode

Snort uses rules stored under: /etc/snort/rules/
and a main configuration file: it depends on the version
/etc/snort/snort.lua (snort3)
/etc/snort/snort.conf (snort2)

4. Testing Snort with a Basic Custom Rule

Create a simple rule file: `nano /etc/snort/rules/local.rules`

Add: `alert icmp any any -> any any (msg:"ICMP test detected"; sid:1000001; rev:1;)`

Save the file.

```
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# ----------------
# LOCAL RULES
# ----------------
# This file intentionally does not come with signatures.  Put your local
# additions here.
#alert icmp any any -> any any (msg: "ICMP Packet Found"; sid:1000001; re
alert icmp any any -> any any (msg:"ICMP test detected"; sid:1000001; rev
~
```

```
ubuntu@ip-10-67-174-238: ~
File  Edit  View  Search  Terminal  Help
''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
        Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
        Copyright (C) 1998-2013 Sourcefire, Inc., et al.
        Using libpcap version 1.9.1 (with TPACKET_V3)
        Using PCRE version: 8.39 2016-06-14
        Using ZLIB version: 1.2.11

        Rules Engine: SF_SNORT_DETECTION_ENGINE  Version 2.4  <Build 1>
        Preprocessor Object: SF_POP  Version 1.0  <Build 1>
        Preprocessor Object: SF_SSH  Version 1.1  <Build 3>
        Preprocessor Object: SF_DNP3  Version 1.1  <Build 1>
        Preprocessor Object: SF_DNS  Version 1.1  <Build 4>
        Preprocessor Object: SF_GTP  Version 1.1  <Build 1>
        Preprocessor Object: SF_MODBUS  Version 1.1  <Build 1>
        Preprocessor Object: SF_SDF  Version 1.1  <Build 1>
        Preprocessor Object: SF_DCERPC2  Version 1.0  <Build 3>
        Preprocessor Object: SF_FTPTELNET  Version 1.2  <Build 13>
        Preprocessor Object: SF_REPUTATION  Version 1.1  <Build 1>
        Preprocessor Object: SF_SMTP  Version 1.1  <Build 9>
```

```
ubuntu@ip-10-67-174-238: ~/Desktop/Task-Exercises
File  Edit  View  Search  Terminal  Help
# unable to create directory '/home//.cache/xdg/dconf': Permission denied.  dcon
f will not work properly.
# unable to create directory '/home//.cache/xdg/dconf': Permission denied.  dcon
f will not work properly.
# Error: Failed to execute child process “tcpreplay” (No such file or directory)
ubuntu@ip-10-67-174-238:~/Desktop/Task-Exercises$ ./traffic-generator.sh
# Option “-e” is deprecated and might be removed in a later version of gnome-ter
minal.
# Use “-- ” to terminate the options and put the command line to execute after i
t.
# unable to create directory '/home//.cache/xdg/dconf': Permission denied.  dcon
f will not work properly.
# unable to create directory '/home//.cache/xdg/dconf': Permission denied.  dcon
f will not work properly.
# unable to create directory '/home//.cache/xdg/dconf': Permission denied.  dcon
f will not work properly.
# Error: Failed to execute child process “tcpreplay” (No such file or directory)
ubuntu@ip-10-67-174-238:~/Desktop/Task-Exercises$ sudo ./traffic-generator.sh
# Option “-e” is deprecated and might be removed in a later version of gnome-ter
minal.
# Use “-- ” to terminate the options and put the command line to execute after i
t.
ubuntu@ip-10-67-174-238:~/Desktop/Task-Exercises$ ^C
ubuntu@ip-10-67-174-238:~/Desktop/Task-Exercises$ sudo ./traffic-generator.sh
```

Run Snort in IDS mode: `sudo snort -c /etc/snort/snort.lua -A (console|full|cmg|none…)`

From your **Metasploitable**, ping the Kali machine. Watch Snort detect it.

Questions

- What alert message appears?

```
          Preprocessor Object: SF_DCERPC2  Version 1.0  <Build 3>
          Preprocessor Object: SF_FTPTELNET  Version 1.2  <Build 13>
          Preprocessor Object: SF_REPUTATION  Version 1.1  <Build 1>
          Preprocessor Object: SF_SMTP  Version 1.1  <Build 9>
          Preprocessor Object: SF_IMAP  Version 1.0  <Build 1>
          Preprocessor Object: SF_SIP  Version 1.1  <Build 1>
          Preprocessor Object: SF_SSLPP  Version 1.1  <Build 4>
ommencing packet processing (pid=1850)
2/14-23:03:55.096196  [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc
ctivity] [Priority: 3] {ICMP} 192.168.175.129 -> 142.250.187.110
2/14-23:03:55.096196  [**] [1:1000001:1] ICMP test detected [**] [Priority: 0]
ICMP} 192.168.175.129 -> 142.250.187.110
2/14-23:03:55.096196  [**] [1:384:5] ICMP PING [**] [Classification: Misc activ
ty] [Priority: 3] {ICMP} 192.168.175.129 -> 142.250.187.110
2/14-23:03:55.409972  [**] [1:1000001:1] ICMP test detected [**] [Priority: 0]
ICMP} 142.250.187.110 -> 192.168.175.129
2/14-23:03:55.409972  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc
activity] [Priority: 3] {ICMP} 142.250.187.110 -> 192.168.175.129
2/14-23:03:55.564684  [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc
ctivity] [Priority: 3] {ICMP} 192.168.175.129 -> 142.250.187.110
2/14-23:03:55.564684  [**] [1:1000001:1] ICMP test detected [**] [Priority: 0]
ICMP} 192.168.175.129 -> 142.250.187.110
2/14-23:03:55.564684  [**] [1:384:5] ICMP PING [**] [Classification: Misc activ
```

- What does **SID** mean in Snort rules?

  Signature ID. ⇒ It is a **unique number** that identifies each Snort rule.

- Why is alerting on ICMP useful (or not)?

  Alerting on ICMP can be useful to detect **reconnaissance**, such as ping sweeps used to discover live hosts.
  However, it is often **not very useful** because ICMP is normally used for legitimate network functions, which can generate **many false alerts**.

Stop Snort with **Ctrl + C**.

Part 3 — Testing Real Snort Attack Rules

5. Enable community rules (already included in Snort on Kali)

List the rules: `ls /etc/snort/rules`

Questions

- How many rule files do you see?

```
ubuntu@ip-10-67-174-238:~/Desktop/Task-Exercises$ ls /etc/snort/rules | wc -l
73
```

- Which ones seem related to web attacks?
  there are to many

```
web-attacks.rules
web-cgi.rules
web-client.rules
web-coldfusion.rules
web-frontpage.rules
web-iis.rules
web-misc.rules
web-php.rules
```

- Which ones seem related to malware or exploits?
  backdoor.rules
  community-exploit.rules
  community-virus.rules
  exploit.rules

6. Detecting a Port Scan

Snort contains rules that detect Nmap scans.
Start Snort: sudo snort -c /etc/snort/snort.conf -A console

From Kali (in a second terminal), run: `nmap -sS <Metasploitable IP>`
Return to Snort's terminal.

Questions

- Which alerts were triggered?
  SYN scan was detected
- Which Snort rule category detected the scan?
  scan.rules
- Why is SYN scan detection important?
  SYN scans are used by attackers to **discover open ports** without completing full TCP connections.

  Detecting them **helps prevent further attacks** by identifying reconnaissance early

Stop Snort.

7. Detecting Web Attacks

Start Snort again.

From Kali, run: `curl http://<Metasploitable IP>:80`

Then try an obvious attack pattern:

```
curl "http://<Metasploitable IP>/index.php?id=1 OR 1=1"
```

Questions

- Which Snort alerts were triggered?

```
curl http:10.10.12.28index.php?id=1 OR 1=1
```

- Did Snort detect potential SQL injection?
  yes
- What signatures or rules were responsible?
  web-application rules specifically SQL injection signatures

Stop Snort.

Part 4 — Writing Advanced Snort Rules

8. Write a rule to detect access to a forbidden directory

Edit local rules: `nano /etc/snort/rules/local.rules`
Add: `alert tcp any any -> any 80 (msg:"Suspicious /admin access"; content:"/admin"; sid:1000002; rev:1;)`
Start Snort.

From Kali: `curl http://<Metasploitable IP>/admin`

Questions

- Did Snort generate an alert?
  Yes, Snort generated an alert when `/admin` was accessed.

  [**] [1:1000002:1] Suspicious /admin access [**]
  [Priority: 0]
  04/15-21:34:12.456789 TCP 192.168.1.50:54321 -> 10.10.12.28:80

- Why is detecting access to /admin directories helpful?
  Because attackers often try `/admin` pages to find hidden or weakly protected admin panels
- How could attackers evade this rule?
  By using URL encoding, changing the path name, using HTTPS, or accessing the admin page through another file or parameter

Stop Snort.

9. Write a rule to detect Telnet usage

Many attacks are done via Telnet due to its lack of encryption.

Add the rule : alert tcp any any -> any 23 (msg:"Telnet connection detected"; sid:1000003; rev:1;)

```
alert icmp any any -> any any (msg: "ICMP Packet Found"; sid:1000001; rev:1;
 alert tcp any any -> any 80 (msg:"Suspicious /admin access"; content:"/admi
sid:1000002; rev:1;)
alert tcp any any -> any 23 (msg:"Telnet connection detected"; sid:1000003;
1;)
```

Start Snort.

From Kali: `telnet <Metasploitable IP>`

Questions

- Did Snort detect the Telnet connection?
  Yes, Snort detected the Telnet connection when traffic on port 23 was generated.
- What information is visible to attackers when Telnet is used?
  Usernames, passwords, and all commands are sent in clear text and can be easily sniffed.
- Which secure alternative should be used instead?
  SSH (Secure Shell)

Stop Snort.

Part 5 — Snort as an IPS (Optional Advanced Section)

Run Snort as an IPS.

```
sudo snort -c /etc/snort/snort.conf -q -Q --daq afpacket
-i eth0:eth1 -A console
```

```
ubuntu@ip-10-64-166-77:~$ sudo snort -c /etc/snort/snort.conf -q -Q --daq afpack
et -i eth0:eth1 -A console

12/15-00:32:30.205660  [**] [1:100000122:1] COMMUNITY WEB-MISC mod_jrun overflov
 attempt [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 10.64
.113.73:47960 -> 10.64.166.77:80
```

Then modify/add one of your local rules to block Telnet.

Questions

- Is the Telnet connection blocked?
  Yes. The IPS blocks (drops) the Telnet traffic when the drop rule is triggered.
- How does IPS differ from IDS?
  IDS: Detects and alerts only (monitoring).

  IPS: Detects **and actively blocks** malicious traffic.

- What are the risks of enabling drop rules on production networks?

  False positives can block legitimate traffic

Service disruption / downtime

Difficult troubleshooting if rules are too strict

Question

Can we detect an exploit (e.g., from Metasploit) using Snort. If yes, how?

Yes, we can detect exploits (for example from Metasploit) using **Snort**.

How?

- Snort uses **signatures (rules)** that match known exploit patterns.
- When Metasploit sends exploit payloads, Snort compares the traffic with its rules.
- If a match is found, Snort generates an **alert** (IDS) or **blocks it** (IPS).